SULIT

# TAKE HOME EXAMINATION: FINAL EXAMINATION
# SESSION 2020/2021, SEMESTER 1

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|---|---|
| **SUBJECT CODE** | **: SCSJ 3323** |
| **SUBJECT NAME** | **: SOFTWARE DESIGN AND ARCHITECTURE** |
| **YEAR/COURSE** | **: 3 SCSJ** |
| **TIME** | **: 5 HOURS** |
| **DATE** | **: 8Th February 2021** |

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**SECTION A: Structured Questions** **(40 Marks)**
**SECTION B: Essay / Problem Solving** **(60 Marks)**

**TOTAL:** **100 MARKS**

**INSTRUCTION AND REMINDER**

Answer all questions in the given answer sheet. All diagrams must be drawn using the UML diagram drawing tools. Note that any form of plagiarism will cause students to get zero in this assessment. Submit the assignment via e-Learning or as instructed by the course lecturer. This examination contributes 35% marks from the whole course assessments.

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|---|---|
| **Name** | |
| **IC No. / Matric No.** | |
| **Year / Programme** | |
| **Section** | |
| **Lecturer Name** | |

**This exam consists of EIGHT (8) pages excluding this page.**

## SECTION A - STRUCTURED QUESTIONS

**INSTRUCTION: Answer all the following FOUR (4) questions.**

The following is the description for **Hotel Ayu Villa System Description document**. This case study will be used throughout questions A1-A4.

**Question A1** **(10marks)**

a) PDL (Programming Design Language) popularity stems from the use of natural languages as opposed to computer languages or graphical techniques to define the required behaviour of functions. Based on the Ayu Villa Hotel System Description document on setting Staff Schedule flowchart written in business process (see section 1.2.1 ) page 6, write up a PDL that shows the business process.

**(4 marks)**

a. The cyclomatic complexity computation allows designers to measure the complexity of flow-based operational designs by determining the complexity of the decision structure of operations instead of lines of code. Based on Figure 1, convert the codes to draw the program flow Graph and determine the cyclomatic complexity using any approach.

```
#include <iostream>
using namespace std;
int main()
{
 char repeat;
 int passenger;
 int age;
double tickectprice;
 do
 {
     cout<<"Please insert the number of passenger(s)"<<endl;
     cin>>passenger;
     tickectprice=0.00;
     for (int a=1;a<=passenger;a++)
     {
      cout<<"Please insert the age for passenger no "<<a<<endl;
         cin>>age;
         if (age<0 || age>100)
         {     cout<<"Please insert the age for passenger no "<<a<<endl;
         cin>>age;
}
         if (age<15)
         {cout<<"Underage is not allowed for this ride. You need to re-
enter again the no. of passenger"<<endl;
         break;}
         tickectprice+=25.00;
     }
     cout<<"Would you like to add passengers? if yes please press
y"<<endl;
     cin>>repeat;
 } while (repeat=='y'||repeat=='Y');
 cout<<"The total price for the ride is RM "<<tickectprice;

      system("pause");
      return 0;}
```

Figure 1

**(6 marks)**

**Question A2** **(10 marks)**

a. Based on the Hotel Ayu Villa System Description document on booking business process in section 1.2.1, draw an EEDT (Extended Entry Decision Table). Give an excerpt of the code for the implementation of the drawn EEDT.

**(10 marks)**

**Question A3** **(10 marks)**

a) An important outcome of the planning stage is to have a clear understanding of the roles that each staff member plays in the process of completing each element of the WBS (Work Breakdown Structure). From your experience in Project 1 with your team members, draw a WBS linear responsibility chart to show distribution of task for development of overall modules based on Hotel Ayu Villa System Description Document.

**(4 marks)**

b) Similar to Gantt charts, network diagrams are also graphical representations of logical flow between tasks, describe task durations, and need the same three basic parameters: Tasks, Tasks durations, and Predecessors and successors. Based on Table 1, draw an Active-on-nodes Network diagram and calculate expected times computed from every path discovered and determine which path is the critical path.

Table 1

| Outline | Time(days) | | | |
|---|---|---|---|---|
| | **Minimum** | **Most likely** | **Maximum** | **Immediate Processors** |
| 1.1 Conduct Interviews | 2 | 2 | 8 | None |
| 1.2 Administer Questionnaires | 2 | 3 | 8 | 1.1 |
| 1.3 Read Company Reports | 2 | 3 | 8 | None |
| 1.4 Analyse data flow | 3 | 7 | 15 | 1.2, 1.3 |
| 1.5 Introduce prototype | 3 | 5 | 8 | 1.2, 1.3 |
| 1.6 Observe reactions to prototypes | 1 | 2 | 9 | 1.5 |
| 1.7 Perform cost benefit analysis | 1 | 2 | 9 | 1.4 |
| 1.8 Prepare proposal | 2 | 2 | 8 | 1.6, 1.7 |
| 1.9 Final review on proposal | 2 | 2 | 8 | 1.8 |
| 1.10 Present Proposal | 2 | 2 | 8 | 1.9 |

**(6 marks)**

**Question A4** **(10 marks)**

a) You are managing a software design project for the given case study composed of 20 tasks. Each task was estimated to cost $500; therefore, BAC = $10,000. Each task was estimated to be completed in two months; therefore, the total planned duration for the project was estimated to be 40 months. Since each task is expected to have the same cost and duration, the completion of an individual task represents 10% completion of the design project. In the current fifth month, only three tasks have been fully completed at a cost of $5,000. You need to provide upper management with current progress status.

Earned Value Management (EVM) is a project management technique to determine the progress of tasks based on the value of the work currently completed versus the work

that was expected to be completed at that particular time. Based on the situation above, show the implementation of the EVM equations in order for you to explain to the upper management.

**(6 marks)**

b) Leadership plays a key role in the success or failure of complex projects. There are **FOUR (4)** keys in leadership skills. Based in your own experience, explain how do you apply **TWO (2)** keys in leadership skills during your project in this course**.**

**(4 marks)**

**SECTION B - ESSAY / PROBLEM SOLVING**

**INSTRUCTION: Answer ALL of the following questions based on the provided case study**

**Question B1** **(30 marks)**

Please answer the following questions based on the **MediNET System Description document**. The following questions describe the additional requirements that require you to apply the suitable design patterns. Each of the following questions does not relate to each other. In your solution (Class Diagram), only draw the modified classes in the existing Package. You do not have to include or redraw all Classes from the Package.

a) In the Billing package described in section 2.1.2, MediNET is required to create a Medical Certificate (MC) if the MC is granted to the patient. There is a new requirement by the paymaster related to the information printed on the MC. All Healthcare Providers (HCPs) agree with three main components to be included in the MC: Header, Body, and Footer. However, they prefer to have different items in each of the main components. Here are the examples of 2 MC settings that can be chosed by HCP:

- MC-Settings-A
    - Header: HCP Name, address, contact number, and MC serial number.
    - Body: Patient name, paymaster name, medical leave start and end dates, and diagnosis information.

- Footer: Medical officer (or doctor) name, isssued date

- MC-Settings-B
  - Header: HCP Logo, HCP Name, address, contact number, and MC serial number.
  - Body: Patient name, paymaster name, and medical leave start and end dates.
  - Footer: Medical officer (or doctor) name, signature, and issued date




**Figure 1**: Examples of different MC formats issued by the HCPs
(only for illustration purpose; they are not related to MC-Settings-A or MC-Settings-B)

Select the most suitable Design Pattern that may solve the additional user requirement above and draw the Class Diagram that applies your selected Design Pattern.

**(10 marks)**

b) Based on the below additional requirement,

In the Registration package described in section 2.1.2, there is a paymaster (PM) that already developed its own class to access its staff information. The PM provides a reusable binary code to access their staff information. Since their staff class contains enough information and no conflicting attribute value for class ID, MediNET team plans to use their distributed binary code. However, instead of using getPatientList(), findPatient(), and getRegNo() as used in the Patient Class, the PM uses getStaffList(), findStaff(), and getStaffID() in its staff class. You are assigned a task to suggest the enhancement of the existing Registration package that complies with the Open-Closed-Principle.

i. Draw a Class Diagram that applies the Adapter Design Pattern to solve the given problem in the new additional requirement. **(5 marks)**

ii. Describe two (2) justifications for how the above solution helps to improve the quality of MediNET design. **(5 marks)**

c) Please read the following additional requirements, select the most suitable design pattern to address this additional requirement, and draw the Class Diagram for your proposed design solution.

MediNET is required to prepare a report of monthly collection from both cash and paymaster patients. These two different types of collections are represented by PayeeCollection and PMCollection classes in Figure 2-2 (see section 2.2.1 of the MediNET System Description document). Select a suitable design pattern that may provide a standardized way for accessing and traversing these different types of collections and draw the related Class Diagram.

**(10 marks)**

**Question B2** **(30 marks)**

Please answer the following questions based on the **Hotel Ayu Villa System Description document**. The following questions describe the additional instructions that require you to apply the suitable design patterns. Each of the following questions does not relate to each other. In your solution (Class Diagram), only draw the modified classes in the existing class diagram. You do not have to include or redraw all Classes.

i. Based on the given case study on Hotel Ayu Villa, choose one scenario that requires you to apply Factory method and draw suitable class diagram based on the scenario.

**(6 marks)**

ii. Give TWO (2) justifications on why the scenario is selected and describe TWO (2) benefits of using that design.

**(4 marks)**

iii. In order to do hotel booking, the system need to check whether it has a previous reservation before the room is available for booking such as dates, make payment, and generate billing refer to Figure 2-4 (see section 2.1.1 of the Hotel Ayu Villa System Description document). Apply Façade design pattern class diagram on the Hotel Ayu Villa booking business process and convert the class diagram and write a code excerpt implementation of the design.

**(10 marks)**

iv. For the promotion notification event scenario under the Customer Services business process (see section 1.2.1 of the Hotel Ayu Villa System Description document) from Hotel Ayu Villa, draw a class diagram of the Observer Design Pattern.

**(10 marks)**